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LAB ASSIGNMENT – 2

Objective:

To understand and implement SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK, retrieve records using SELECT queries, and perform INSERT, UPDATE, and ALTER operations on tables.

Table Structures:

CLASS

D (NUMBER) and Name (VARCHAR).

The initial records in the table are (1, 'Bravo'), (2, 'Alex'), and (4, 'Cheng').

EMPLOYEES

Employee_Id (NUMBER), First_Name (VARCHAR), Last_Name (VARCHAR), Email (VARCHAR), Phone_Number (VARCHAR), Hire_Date (DATE), Job_Id (VARCHAR), Salary (NUMBER), Commission_Pct (NUMBER), Manager_Id (NUMBER), and Department_Id (NUMBER).

DDL Questions (Data Definition Language):

1. How do you create a table named CLASS with appropriate data types?

STEP 1 – FIRST WE CREATE DATABASE .

```
CREATE DATABASE HARSH_BARDHAN;
USE HARSH_BARDHAN ;
```

STEP 2 – THEN WE CREATE TABLE.

```
CREATE TABLE CLASS(
ID INT,
NAME VARCHAR(100)
);
```

STEP 3 – THEN WE INSERT DATA INTO THE TABLE.

```
INSERT INTO CLASS VALUES (1, 'Bravo');
INSERT INTO CLASS VALUES (2, 'Alex');
INSERT INTO CLASS VALUES (4, 'Cheng');
COMMIT;
SELECT * FROM CLASS;
```

OUTPUT:

Result Grid

ID	NAME
1	Bravo
2	Alex
4	Cheng

2.) How do you create a table named EMPLOYEES with the specified attributes?

```
CREATE TABLE EMPLOYEES (
    Employee_Id INT PRIMARY KEY,
    First_Name VARCHAR(30),
    Last_Name VARCHAR(30),
    Email VARCHAR(50),
    Phone_Number VARCHAR(20),
    Hire_Date DATE,
    Job_Id VARCHAR(20),
    Salary INT ,
    Commission_Pct INT ,
    Manager_Id INT ,
    Department_Id INT
);
SELECT* FROM EMPLOYEES;
```

OUTPUT:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3.) How do you create a table named Client_master with the specified attributes?

```sql
CREATE TABLE Client_master (
    ClientNO VARCHAR(10) PRIMARY KEY,
    Name VARCHAR(30),
    City VARCHAR(30),
    bal_due INT
);
SELECT* FROM Client_master;
```

OUTPUT :

Result Grid | Filter Rows:

	ClientNO	Name	City	bal_due
*	NULL	NULL	NULL	NULL

4.) What SQL statement would you use to rename the table Client_master to Client12?

```sql
ALTER TABLE Client_master RENAME TO Client12;
```

5.) How do you display the bal_due column heading as "BALANCE"?

```sql
SELECT bal_due AS "BALANCE"
FROM Client12;
```

OUTPUT :

Result Grid | Filter Rows:

	BALANCE

DML Questions (Data Manipulation Language):

Part A: Transactions and Savepoints (CLASS Table)

Give the output of the following SQL script:

INSERT INTO class VALUES (5,'Rahul');

COMMIT;

UPDATE class SET name = 'Abhijeet' WHERE id='5';

SAVEPOINT A;

INSERT INTO class VALUES (6,'Chris');

SAVEPOINT B;

INSERT INTO class VALUES (7,'Bravo');

SAVEPOINT C;

SELECT * FROM class;

ROLLBACK TO B;

SELECT * FROM class;

ROLLBACK TO A;

i) INSERT INTO class VALUES (5,'Rahul');

COMMIT;

```
INSERT INTO CLASS VALUES (5, 'Rahul');
COMMIT;
SELECT * FROM CLASS;
```

Result Grid

	ID	NAME
▶	1	Bravo
	2	Alex
	4	Cheng
	5	Rahul

ii) UPDATE class SET name = 'Abhijeet' WHERE id='5';

```
SET AUTOCOMMIT = 0;
SET SQL_SAFE_UPDATES = 0;
UPDATE class SET name = 'Abhijeet' WHERE id='5';
```

<u>OUTPUT :</u>

Result Grid

	ID	NAME
▶	1	Bravo
	2	Alex
	4	Cheng
	5	Abhijeet

iii) SAVEPOINT A;

```
SAVEPOINT A;
```

iv) INSERT INTO class VALUES (6,'Chris');

SAVEPOINT B;

```
INSERT INTO CLASS VALUES (6, 'Chris');
SAVEPOINT B;
```

OUTPUT :

Result Grid

	ID	NAME
▶	1	Bravo
	2	Alex
	4	Cheng
	5	Abhijeet
	6	Chris

v) INSERT INTO class VALUES (7,'Bravo');

SAVEPOINT C;

```
INSERT INTO CLASS VALUES (7, 'Bravo');
SAVEPOINT C;
SELECT * FROM CLASS;
```

OUTPUT :

Result Grid

	ID	NAME
▶	1	Bravo
	2	Alex
	4	Cheng
	5	Abhijeet
	6	Chris
	7	Bravo

vi) ROLLBACK TO B;

SELECT * FROM class;

```
ROLLBACK TO B;
SELECT * FROM CLASS;
```

OUTPUT :

Result Grid

	ID	NAME
▶	1	Bravo
	2	Alex
	4	Cheng
	5	Abhijeet
	6	Chris
	7	Bravo
	6	Chris

vii) ROLLBACK TO A;

```
ROLLBACK TO A;
SELECT* FROM CLASS;
```

OUTPUT :

Result Grid

	ID	NAME
▶	1	Bravo
	2	Alex
	4	Cheng
	5	Abhijeet

Part B: Retrieving Records (EMPLOYEES Table)

1.) Find out the employee id, names, and salaries of all the employees..

```
INSERT INTO EMPLOYEES VALUES
(101,'KRISH','SINGHAL','krish@gmail.com','9876543210','2023-01-10','SOFTWARE_DEV',12000,NULL,100,60);
INSERT INTO EMPLOYEES VALUES
(102,'HARSH','VARDHAN','harsg@gmail.com','9876543211','2023-02-12','HR_REP',4800,NULL,101,70);
INSERT INTO EMPLOYEES VALUES
(103,'HRIDYANSHU','SINGH','hridyanshu@gmail.com','9876543212','2023-03-15','SA_REP',5200,0.10,100,80);
INSERT INTO EMPLOYEES VALUES
(104,'ANKIT','SINGH','abkit@gmail.com','9876543213','2023-04-18','SOFTWARE_DEV',4500,NULL,102,60);
INSERT INTO EMPLOYEES VALUES
(105,'NIRVIKALP','CHAIDHARy','nirvikalp@gmail.com','9876543214','2023-05-20','SA_REP',7000,0.15,100,80);
INSERT INTO EMPLOYEES VALUES
(106,'ROHAN','YADAV','rohan@gmail.com','9876543215','2023-06-22','HR_REP',5000,NULL,103,70);
COMMIT;
-- Employee id, name & salary
SELECT Employee_Id, First_Name, Last_Name, Salary
FROM EMPLOYEES;
```

OUTPUT :

Employee_Id	First_Name	Last_Name	Salary
101	KRISH	SINGHAL	12000
102	HARSH	VARDHAN	4800
103	HRIDYANSHU	SINGH	5200
104	ANKIT	SINGH	4500
105	NIRVIKALP	CHAIDHARy	7000
106	ROHAN	YADAV	5000
NULL	NULL	NULL	NULL

2.) List out the employees who work under manager 100.

```
SELECT *
FROM EMPLOYEES
WHERE Manager_Id = 100;
```

OUTPUT:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	KRISH	SINGHAL	krish@gmail.com	9876543210	2023-01-10	SOFTWARE_DEV	12000	NULL	100	60
103	HRIDYANSHU	SINGH	hridyanshu@gmail.com	9876543212	2023-03-15	SA_REP	5200	0.1	100	80
105	NIRVIKALP	CHAIDHARy	nirvikalp@gmail.com	9876543214	2023-05-20	SA_REP	7000	0.15	100	80
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3.) Find the names of the employees who have a salary greater than or equal to 4800.

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Salary >= 4800;
```

OUTPUT :

First_Name	Last_Name
KRISH	SINGHAL
HARSH	VARDHAN
HRIDYANSHU	SINGH
NIRVIKALP	CHAIDHARy
ROHAN	YADAV

4.) List out the employees whose last name is 'AUSTIN'.

```
SELECT *
FROM EMPLOYEES
WHERE Last_Name = 'Austin';
```

OUTPUT :

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5.) Find the names of the employees who work in departments 60, 70, and 80.

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Department_Id IN (60,70,80);
```

OUTPUT :

First_Name	Last_Name
KRISH	SINGHAL
HARSH	VARDHAN
HRIDYANSHU	SINGH
ANKIT	SINGH
NIRVIKALP	CHAIDHARy
ROHAN	YADAV

6.) Display the unique Manager_Id.

```
SELECT DISTINCT Manager_Id
FROM EMPLOYEES;
```

OUTPUT :

Result Grid

Manager_Id
100
101
102
103

Part C: Updating Records (Client_master Table)

1.) Insert five records into the Client_master table.

```sql
INSERT INTO Client12 VALUES ('C123','Aryan Sharma','Delhi',4500);
INSERT INTO Client12 VALUES ('C124','Anita Verma','Mumbai',6000);
INSERT INTO Client12 VALUES ('C125','Kaira Singh','Jaipur',7500);
INSERT INTO Client12 VALUES ('C126','Harsh Mehta','Noida',3000);
INSERT INTO Client12 VALUES ('C127','Ashu Joshi','Chandigarh',8200);
```

OUTPUT :

ClientNO	Name	City	bal_due
C123	Aryan Sharma	Delhi	4500
C124	Anita Verma	Mumbai	6000
C125	Kaira Singh	Jaipur	7500
C126	Harsh Mehta	Noida	3000
C127	Ashu Joshi	Chandigarh	8200
NULL	NULL	NULL	NULL

2.) Find the names of clients whose bal_due > 5000.

```
SELECT Name
FROM Client12
WHERE bal_due > 5000;
```

OUTPUT :

Name
Anita Verma
Kaira Singh
Ashu Joshi

3.) Change the bal_due of ClientNO "C123" to Rs. 5100.

```
UPDATE Client12
SET bal_due = 5100
WHERE ClientNO = 'C123';
COMMIT;
SELECT* FROM Client12;
```

OUTPUT :

ClientNO	Name	City	bal_due
C123	Aryan Sharma	Delhi	5100
C124	Anita Verma	Mumbai	6000
C125	Kaira Singh	Jaipur	7500
C126	Harsh Mehta	Noida	3000
C127	Ashu Joshi	Chandigarh	8200
NULL	NULL	NULL	NULL